



DEPARTMENT OF PHYSICS

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Master's Thesis

Simulation of Radiative Effects for the AMBER Proton Radius Measurement with McMule

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Abstract

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1 Introduction

The proton is one of the basic building blocks of ordinary matter and, at the same time, a highly complex system. It is a composite particle made of quarks and gluons, with a valence-quark content of two up quarks and one down quark, bound together by the strong interaction. While its constituents and their interactions are described by quantum chromodynamics (QCD), many properties of the proton cannot be obtained straightforwardly from perturbative calculations. Precise measurements of its electromagnetic properties therefore provide valuable information about its structure and are essential for testing our theoretical understanding of the proton.

Since the electric charge of the proton is distributed over a finite volume rather than concentrated at a single point, its spatial extent can be characterized by the proton charge radius. The electromagnetic structure of the proton is described by form factors, which parameterize its charge and magnetization distributions. In particular, the electric Sachs form factor $G_E(Q^2)$ describes the electric structure of the proton as a function of the squared four-momentum transfer Q^2 . Its slope at zero momentum transfer is directly related to the proton charge radius,

$$r_p^2 = -6 \left. \frac{dG_E(Q^2)}{dQ^2} \right|_{Q^2=0}.$$

A precise determination of the proton charge radius therefore requires measurements at very small momentum transfers and an accurate extrapolation of the electric form factor towards $Q^2 = 0$.

Experimentally, the proton charge radius can be determined using two different approaches. The first is based on spectroscopy, in which atomic energy differences are determined by measuring the absorption or emission of electromagnetic radiation. A particularly sensitive observable is the Lamb shift. In the Dirac description of hydrogen with a point-like proton, the $2S_{1/2}$ and $2P_{1/2}$ states are degenerate. However, in 1947, Lamb and Retherford experimentally demonstrated that the $2S_{1/2}$ state lies slightly above the $2P_{1/2}$ state in energy. This observation played an important role in the development of quantum electrodynamics (QED), which explains most of this energy difference through radiative corrections.

The proton charge radius introduces an additional contribution to these atomic energy levels. This effect is especially important for S -states, whose wave functions have a non-zero probability density at the position of the proton and therefore overlap with its charge distribution. The $2P$ -state, in contrast, has a vanishing probability density at the origin and is affected much less. As a result, the measured energy difference between the $2S$ and $2P$ states also contains information about the proton charge radius. By comparing precisely measured transition frequencies with theoretical predictions, the proton radius can therefore be extracted.

Measurements of the Lamb shift in muonic hydrogen provide an especially precise determination of the proton radius. Muonic hydrogen is a hydrogen-like atom in which the electron is replaced by a negatively charged muon. Due to the much larger mass of the muon compared to the electron, its Bohr radius is approximately 200 times smaller than that of ordinary hydrogen. As a result, the muon has a significantly larger probability density close to the proton, making the energy levels much more sensitive to the finite proton size.

The second approach uses elastic lepton–proton scattering experiments, where the differential scattering cross section provides access to the proton’s electromagnetic form factors. More details on this approach will be discussed in Chapter 3.

Historically, measurements based on elastic electron–proton scattering and spectroscopy of ordinary hydrogen yielded a proton charge radius of approximately 0.88 fm. This picture changed significantly in 2010, when a high-precision measurement of the Lamb shift in muonic hydrogen led to a significantly smaller value of approximately 0.84 fm. The discrepancy between this result and previous determinations became known as the proton radius puzzle. Figure 1.1 illustrates this discrepancy and summarizes several experimental determinations of the proton charge radius.

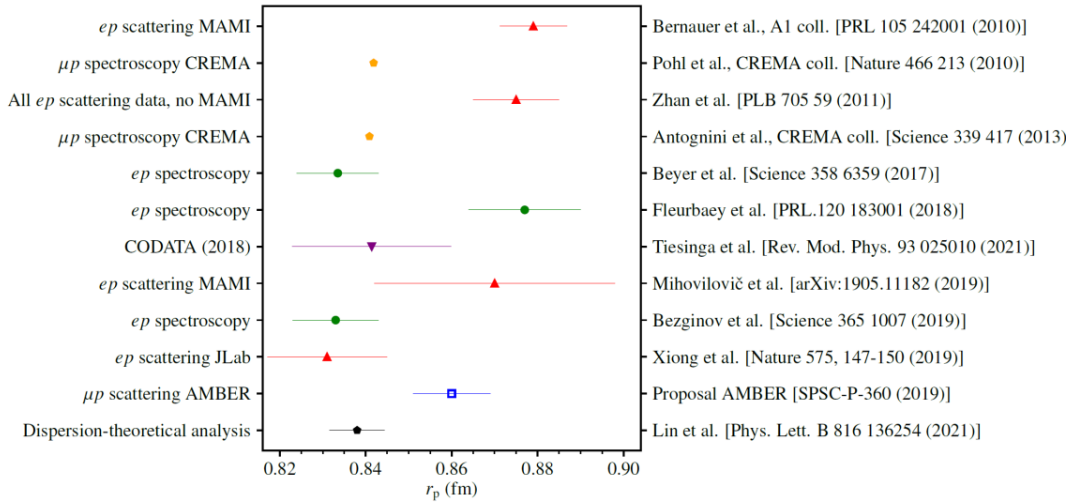


Figure 1.1: Comparison of proton charge radius measurements from different experiments. Figure taken from [Karl PAW25].

Although more recent measurements have substantially reduced the discrepancy, an independent high-precision determination of the proton radius remains important. In this context, elastic muon–proton scattering provides such an independent approach. A new precision measurement using this method is pursued by the AMBER experiment at CERN, which will be presented in detail in Chapter 2.

2 The AMBER Experiment

2.1 Overview of the AMBER Experiment

The AMBER (Apparatus for Meson and Baryon Experimental Research) experiment, designated NA66, is a fixed-target experiment located in the EHN2 experimental hall in the North Area of CERN, as shown in Figure 2.1. It uses the M2 beam line of the Super Proton Synchrotron (SPS), which provides high-energy muon and hadron beams to the experiment.

AMBER builds on the infrastructure and large-acceptance spectrometer of the former COMPASS experiment and extends it with detector systems developed for its new physics program. Its measurements cover several topics in hadron physics, including hadron structure, spectroscopy, and a high-precision determination of the proton charge radius. For the proton-radius measurement, AMBER studies muon-proton scattering at very small four-momentum transfers $1 \times 10^{-3} \text{ GeV}^2 < Q^2 < 4 \times 10^{-2} \text{ GeV}^2$, where the electric form factor is particularly sensitive to the proton charge radius.

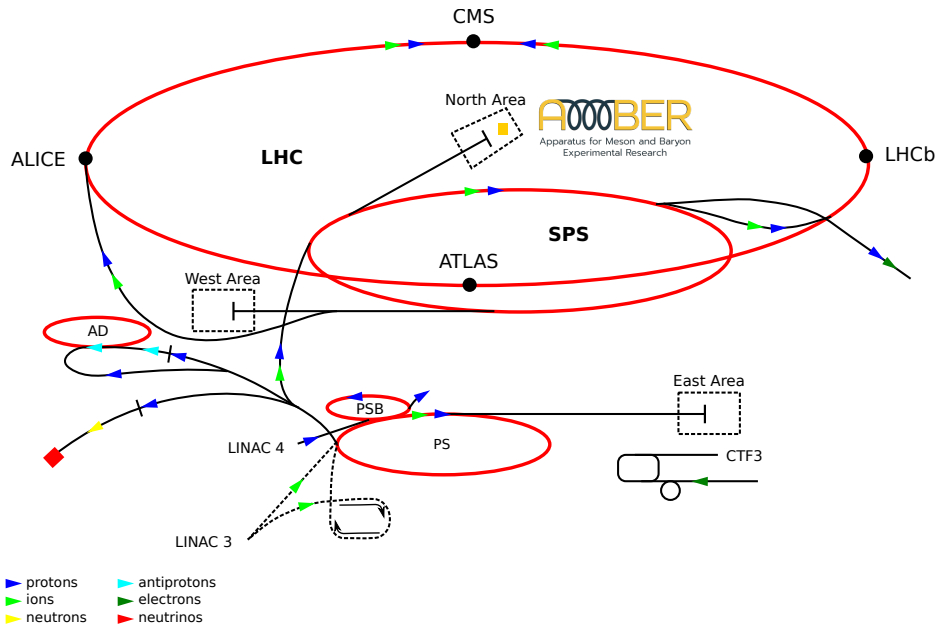


Figure 2.1: Location of the AMBER experiment within the CERN accelerator complex. Figure adapted from [1].

2.2 Experimental Setup for the Proton Radius Measurement

The proton-radius measurement is performed using a 100 GeV muon beam. The beam is delivered to the detector in spills. A spill refers to a single period during which the beam is extracted from the accelerator and delivered to the experiment. For AMBER, a spill is approximately 4.8 s long. Therefore, an intensity of approximately 4×10^6 muons per spill corresponds to an average rate of about 8.3×10^5 muons per second that hit the target during the spill.

The beam is directed onto an active hydrogen target operated at a pressure of 20 bar. The high gas pressure increases the target density and therefore the probability of muon–proton interactions. Since hydrogen provides a clean proton target, it allows the proton kinematics to be reconstructed without additional nuclear effects.

The recoil proton is reconstructed inside the target using a Time Projection Chamber (TPC), while the incoming and scattered muons are measured by dedicated tracking detectors and the downstream spectrometer. An overview of the experimental setup is shown in Figure 2.2.

The direction of the incoming beam defines the longitudinal axis of the experiment. Detector components located before the interaction point with respect to the beam direction are referred to as upstream, whereas components located after the interaction point are called downstream.

The main detector components relevant for the proton-radius measurement are described in the following.

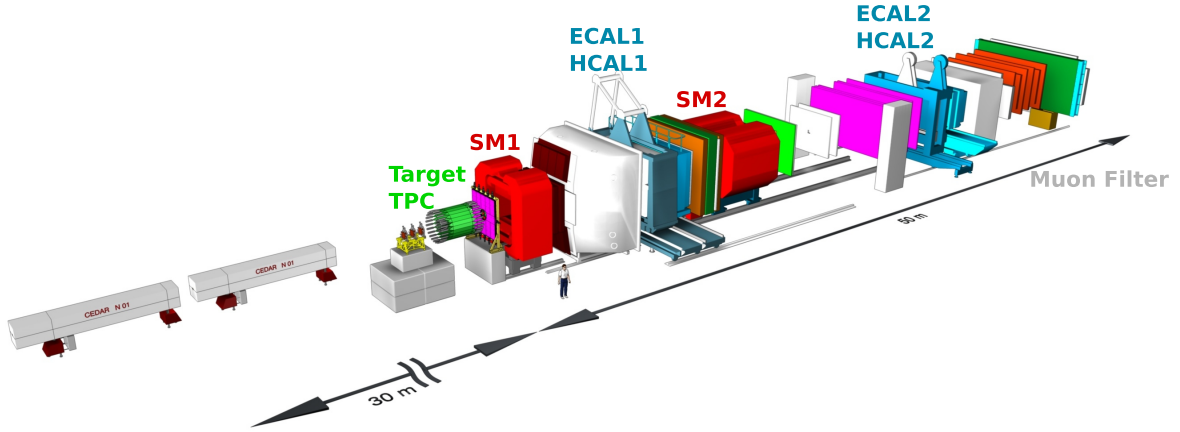


Figure 2.2: Overview of the AMBER setup, with the detector components relevant for the proton-radius measurement labeled. Figure adapted from [XXX].

Unified Tracking Stations Four Unified Tracking Stations (UTS) surround the target region, with two stations located upstream and two downstream of the TPC. Their main purpose is to reconstruct the trajectories of the incoming and scattered muons and thereby determine the small muon scattering angles with high precision.

Each UTS combines Silicon Pixel Detectors (SPD) based on ALPIDE Monolithic Active Pixel Sensors (MAPS) with Scintillating Fiber Hodoscopes (SFH). The ALPIDE sensors provide

the high spatial resolution required for precise track reconstruction. Their small pixel size of approximately $28 \times 28 \mu\text{m}^2$ allows the position of the traversing muon to be measured accurately.

The SFH complement the pixel detectors by providing precise timing information. This is particularly important because the comparatively long integration time of the ALPIDE sensors can result in hits from several beam particles being recorded within the same time window. The time information provided by the SFH allows these ambiguities to be resolved and helps to associate the reconstructed muon tracks with the corresponding event in the TPC.

Time Projection Chamber The TPC serves as an active hydrogen target. It is operated with hydrogen gas at a pressure of 20 bar and is divided into two drift cells, each with a length of 40 cm. In total, the system contains approximately 20 m^3 of hydrogen. Thin carbon-fiber composite windows are used where the muon beam enters and leaves the TPC in order to minimize interactions of the beam particles with detector material.

When a charged recoil proton traverses the hydrogen gas, it ionizes the gas molecules and creates electron-ion pairs along its trajectory. A high voltage of approximately 100 kV generates the electric drift field, causing the liberated electrons to drift towards the readout plane. A Frisch grid is placed in front of the anode to shield it from the drifting electrons until they reach the readout region. This reduces the dependence of the measured signal on the position at which the ionization was produced and improves the energy and time resolution.

The anode is segmented into pads, allowing the position and arrival time of the electrons to be measured and the three-dimensional proton track to be reconstructed.

In addition to the track direction, the energy deposited by the proton provides information about its kinetic energy. For an initial proton at rest, the squared four-momentum transfer is directly related to the recoil-proton kinetic energy. The TPC therefore provides direct access to the recoil-proton kinematics and to the momentum transfer of the scattering process.

To maintain stable detector operation, the hydrogen is continuously recirculated and filtered, keeping impurity levels below approximately 1 ppm.

Spectrometer Magnets Downstream of the target region, the AMBER spectrometer contains two dipole magnets, SM1 and SM2. For the proton-radius measurement, the first magnet, SM1, is switched off, while the second dipole magnet, SM2, is operated. The deflection in SM2, together with downstream tracking detectors such as the GEM stations, allows the momentum of the scattered muon to be reconstructed.

Since elastic muon-proton scattering in the investigated low- Q^2 region is strongly forward-peaked, precise reconstruction of the small scattering angles and the high-momentum outgoing muon is essential.

Electromagnetic and Hadronic Calorimeters The AMBER spectrometer features a two-stage calorimeter system containing two electromagnetic calorimeters (ECAL1 and ECAL2) and

two hadronic calorimeters (HCAL1 and HCAL2). The calorimeters measure the energy of particles through the detection of the energy deposited in active detector material.

Electromagnetic calorimeters are primarily sensitive to electrons and photons, which produce electromagnetic showers in the detector material, whereas hadronic calorimeters are designed to measure strongly interacting particles.

For the proton-radius measurement, ECAL2 is particularly important because it provides sensitivity to photons emitted in radiative muon–proton scattering. Such photon emission modifies the kinematics of the observed final state and therefore has to be taken into account in the determination of radiative corrections. Although ECAL2 did not provide data for the proton-radius run before Long Shutdown 3, future measurements will be important for comparisons with theoretical predictions and the simulation framework developed in this thesis.

Muon Filters Muon identification is provided by the downstream AMBER spectrometer. Dedicated muon-filter systems consist of thick absorber material followed by tracking detectors. Hadrons and electromagnetic shower components are strongly attenuated in the absorber, whereas high-energy muons can penetrate the material. Tracks detected downstream of the absorber can therefore be matched to reconstructed spectrometer tracks to identify the scattered muon.

3 Theoretical Background on Elastic Muon–Proton Scattering

Throughout this chapter, natural units with $\hbar = c = 1$ and the metric convention

$$g^{\mu\nu} = \text{diag}(1, -1, -1, -1)$$

are used.

3.1 From Transition Rates to Scattering Cross Sections

The transition rate of a quantum mechanical process is described by Fermi's golden rule. It relates the probability of a transition between an initial and a final state to the interaction matrix element and the available phase space,

$$\Gamma_{i \rightarrow f} = 2\pi |\mathcal{M}_{fi}|^2 \rho(E_f). \quad (3.1)$$

Here, $\mathcal{M}_{fi}$ is the transition matrix element and contains the dynamical information about the interaction. The factor $\rho(E_f)$ describes the density of accessible final states and therefore contains the kinematic information of the process. It depends on the masses, energies, and momenta of the particles involved.

In relativistic quantum field theory, the corresponding quantity is the invariant scattering amplitude $\mathcal{M}$, which is obtained from the relevant Feynman diagrams. The nonrelativistic transition matrix element $\mathcal{M}_{fi}$ and the relativistic invariant amplitude $\mathcal{M}$ play similar roles, but are defined using different normalization conventions for the particle states.

For the decay of one initial particle with four-momentum p_1 into n final-state particles, the differential transition rate is given by

$$d\Gamma = \frac{(2\pi)^4}{2E_1} |\mathcal{M}|^2 \delta^{(4)}\left(p_1 - \sum_f p_f\right) d\Phi_n, \quad (3.2)$$

where the Lorentz-invariant phase-space element is

$$d\Phi_n = \prod_f \frac{d^3\mathbf{p}_f}{(2\pi)^3 2E_f}. \quad (3.3)$$

The four-dimensional delta function ensures conservation of energy and momentum. For scattering processes, the relevant observable is not the transition rate itself but the cross section. It is obtained by normalizing the transition rate to the incident particle flux,

$$d\sigma = \frac{d\Gamma}{\text{flux}}. \quad (3.4)$$

The flux factor accounts for the relative motion of the incoming particles and normalizes the transition probability to the number of possible collisions. For a general $2 \rightarrow n$ scattering process, the Lorentz-invariant cross section is

$$d\sigma = \frac{(2\pi)^4 |\overline{\mathcal{M}}|^2}{4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}} \delta^{(4)}\left(p_1 + p_2 - \sum_f p_f\right) d\Phi_n. \quad (3.5)$$

For an unpolarized process, the bar denotes an average over the initial-state spins and a sum over the final-state spins.

3.2 Non-Radiative Kinematics

The starting point for the following discussion is the non-radiative elastic muon–proton scattering process

$$p(p_1) + \mu(p_2) \rightarrow p(p_3) + \mu(p_4). \quad (3.6)$$

In the laboratory frame, the initial proton is at rest. The incoming four-momenta are therefore

$$p_1 = (m_p, \mathbf{0}), \quad p_2 = (E_\mu, \mathbf{p}_\mu), \quad (3.7)$$

where m_p is the proton mass and E_μ is the energy of the incoming muon. The outgoing proton and muon are described by

$$p_3 = (E'_p, \mathbf{p}'_p), \quad p_4 = (E'_\mu, \mathbf{p}'_\mu). \quad (3.8)$$

Energy and momentum conservation require

$$p_1 + p_2 = p_3 + p_4. \quad (3.9)$$

At leading order, the interaction is mediated by the exchange of a virtual photon with four-momentum

$$q = p_2 - p_4 = p_3 - p_1. \quad (3.10)$$

The invariant momentum transfer is defined as

$$Q^2 = -q^2, \quad (3.11)$$

such that $Q^2 > 0$ for the spacelike momentum transfers relevant to elastic scattering. The value of Q^2 determines the spatial resolution of the interaction: larger momentum transfers correspond to smaller distance scales being probed inside the proton.

The Mandelstam variables provide a convenient Lorentz-invariant description of the scattering kinematics. For the elastic $2 \rightarrow 2$ process considered here, they are

$$\begin{aligned} s &= (p_1 + p_2)^2 = (p_3 + p_4)^2 = m_p^2 + m_\mu^2 + 2m_p E_\mu, \\ t &= (p_1 - p_3)^2 = (p_2 - p_4)^2 = 2m_p^2 - 2m_p E'_p = -Q^2, \end{aligned} \quad (3.12)$$

and

$$u = (p_1 - p_4)^2 = (p_2 - p_3)^2 = m_p^2 + m_\mu^2 - 2m_p E'_\mu.$$

They satisfy

$$s + t + u = 2m_p^2 + 2m_\mu^2.$$

3.3 Differential Cross Section for Elastic Muon–Proton Scattering

The differential cross section for elastic muon–proton scattering can be expressed in terms of the Mandelstam variables and the invariant matrix element. For a general $2 \rightarrow 2$ process,

$$\frac{d\sigma}{dt} = \frac{1}{64\pi s p_{\text{cm}}^2} |\overline{\mathcal{M}}|^2,$$

where p_{cm} denotes the magnitude of either incoming particle momentum in the center-of-mass frame. It can be expressed using the Källén function,

$$\lambda(s, m_\mu^2, m_p^2) = s^2 + m_\mu^4 + m_p^4 - 2s(m_\mu^2 + m_p^2) - 2m_\mu^2 m_p^2,$$

through

$$p_{\text{cm}}^2 = \frac{\lambda(s, m_\mu^2, m_p^2)}{4s}.$$

The differential cross section can therefore be written in the Lorentz-invariant form

$$\frac{d\sigma}{dt} = \frac{1}{16\pi \lambda(s, m_\mu^2, m_p^2)} |\overline{\mathcal{M}}|^2. \quad (3.13)$$

While $d\sigma/dt$ provides a compact invariant description, the experimentally accessible quantities are the energy and scattering angle of the outgoing muon. It is therefore useful to express the cross section in terms of the solid angle,

$$\frac{d\sigma}{d\Omega} = \frac{d\sigma}{dt} \left| \frac{dt}{d\Omega} \right|. \quad (3.14)$$

Since $t = -Q^2$, see Equation 3.12, the transformation factor becomes

$$\left| \frac{dt}{d\Omega} \right| = \frac{dQ^2}{d\Omega}. \quad (3.15)$$

No massless approximation is made for the muon at this stage. In the laboratory frame, the magnitudes of the incoming and outgoing muon three-momenta are

$$p_\mu = \sqrt{E_\mu^2 - m_\mu^2}, \quad p'_\mu = \sqrt{E_\mu'^2 - m_\mu^2}. \quad (3.16)$$

The outgoing proton is a real, on-shell particle and therefore satisfies $p_3^2 = m_p^2$. Since $p_3 = p_1 + q$, this condition gives

$$Q^2 = 2m_p(E_\mu - E'_\mu). \quad (3.17)$$

Independently, evaluating $Q^2 = -(p_2 - p_4)^2$ gives

$$Q^2 = 2(E_\mu E'_\mu - p_\mu p'_\mu \cos \theta_\mu - m_\mu^2), \quad (3.18)$$

where θ_μ is the laboratory scattering angle of the outgoing muon. Equations (3.17) and (3.18) determine the final-state energy and momentum transfer for a given incoming energy and scattering angle. Solving for the physical outgoing-muon energy gives

$$E'_\mu = \frac{(m_p + E_\mu)(m_p E_\mu + m_\mu^2) + p_\mu^2 \cos \theta_\mu \sqrt{m_p^2 - m_\mu^2 \sin^2 \theta_\mu}}{(m_p + E_\mu)^2 - p_\mu^2 \cos^2 \theta_\mu}. \quad (3.19)$$

For an azimuthally symmetric unpolarized cross section, the azimuthal angle can be integrated from 0 to 2π , such that

$$d\Omega = 2\pi \sin \theta_\mu d\theta_\mu. \quad (3.20)$$

The corresponding transformation is therefore

$$\frac{dQ^2}{d\Omega} = \frac{1}{2\pi \sin \theta_\mu} \left| \frac{dQ^2}{d\theta_\mu} \right|.$$

Differentiating the exact kinematic relations gives

$$\frac{dQ^2}{d\theta_\mu} = \frac{2m_p p_\mu (p'_\mu)^2 \sin \theta_\mu}{p'_\mu (m_p + E_\mu) - p_\mu E'_\mu \cos \theta_\mu}. \quad (3.21)$$

Consequently,

$$\frac{dQ^2}{d\Omega} = \frac{m_p p_\mu (p'_\mu)^2}{\pi [p'_\mu (m_p + E_\mu) - p_\mu E'_\mu \cos \theta_\mu]}. \quad (3.22)$$

In the laboratory frame, the Källén function satisfies

$$\lambda(s, m_\mu^2, m_p^2) = 4m_p^2 p_\mu^2.$$

Inserting this relation together with Equation 3.22 into Equation 3.13 gives

$$\frac{d\sigma}{d\Omega} = \frac{(p'_\mu)^2}{64\pi^2 m_p p_\mu \left[p'_\mu (m_p + E_\mu) - p_\mu E'_\mu \cos \theta_\mu \right]} |\overline{\mathcal{M}}|^2. \quad (3.23)$$

This expression contains the exact two-body phase-space and recoil factors for a finite muon mass. The remaining task is to determine the matrix element $\mathcal{M}$ describing the electromagnetic interaction.

3.4 Born Approximation

The starting point for calculating the scattering amplitude is the interaction part of the QED Lagrangian. At leading order in quantum electrodynamics, referred to as the Born approximation, muon–proton scattering is described by the exchange of a single virtual photon.

For a charged fermion, the electromagnetic interaction is described by

$$\mathcal{L}_{\text{int}} = -e Q_f \bar{\psi} \gamma^\mu \psi A_\mu,$$

where ψ is the Dirac spinor field representing the charged fermion, A_μ is the electromagnetic four-potential, γ^μ are the Dirac gamma matrices, $e > 0$ is the elementary charge, and Q_f is the fermion charge in units of e .

Introducing the electromagnetic current

$$j_{\text{em}}^\mu = Q_f \bar{\psi} \gamma^\mu \psi,$$

the interaction Lagrangian becomes

$$\mathcal{L}_{\text{int}} = -e j_{\text{em}}^\mu A_\mu.$$

This expression directly shows that the electromagnetic interaction arises from the coupling between the electromagnetic current and the electromagnetic field. In a scattering process, this interaction is represented diagrammatically by a vertex connecting a charged fermion to a photon.

For a point-like fermion such as the muon, the electromagnetic vertex follows directly from the QED interaction Lagrangian. Apart from the charge factor Q_μ , the corresponding Feynman rule is

$$-ie\gamma^\mu.$$

Including the incoming and outgoing muon spinors, the muon vertex is

$$Q_\mu \bar{u}(p_4) (-ie\gamma^\mu) u(p_2),$$

where $u(p_2)$ and $\bar{u}(p_4)$ describe the incoming and outgoing muon, respectively.

For a composite particle such as the proton, the electromagnetic vertex is modified by its internal structure. This modification is described by electromagnetic form factors, which characterize the proton's internal charge and magnetization structure as a function of the momentum transfer. The proton vertex can be written as

$$\bar{u}(p_3)(-ie\Gamma^\nu(Q^2))u(p_1),$$

with the vertex function

$$\Gamma^\nu(Q^2) = F_1(Q^2)\gamma^\nu + \frac{i\sigma^{\nu\rho}q_\rho}{2m_p}F_2(Q^2), \quad (3.24)$$

where

$$\sigma^{\nu\rho} = \frac{i}{2}[\gamma^\nu, \gamma^\rho].$$

The Dirac form factor F_1 describes the charge-like part of the proton's electromagnetic interaction, while the Pauli form factor F_2 accounts for its anomalous magnetic contribution. At zero momentum transfer, they satisfy

$$F_1(0) = 1, \quad F_2(0) = \kappa_p = \mu_p - 1,$$

where κ_p is the anomalous magnetic moment and $\mu_p = 2.792847$ is the dimensionless proton magnetic moment in units of the nuclear magneton.

The Q^2 dependence of the two proton form factors is illustrated in Figure 3.1. Both form factors change as the momentum transfer increases, reflecting the fact that the proton is not point-like. The comparison of the two calculation schemes also illustrates how different theoretical treatments of the hadronic structure affect the detailed Q^2 dependence.

For the interpretation of elastic scattering, it is useful to introduce the Sachs form factors. They are related to the Dirac and Pauli form factors by

$$G_E(Q^2) = F_1(Q^2) - \frac{Q^2}{4m_p^2}F_2(Q^2) \quad (3.25)$$

and

$$G_M(Q^2) = F_1(Q^2) + F_2(Q^2). \quad (3.26)$$

Their normalizations at zero momentum transfer are

$$G_E(0) = 1, \quad G_M(0) = \mu_p.$$

The Sachs form factors provide a more direct description of the electric and magnetic structure of the proton. Their Q^2 dependence is shown in Figure 3.2. Both G_E and G_M decrease as the momentum transfer increases, indicating that the electromagnetic structure of the proton becomes resolved. Of particular importance for the proton-radius measurement is the behavior of G_E close to $Q^2 = 0$. As discussed in the introduction, the slope of G_E in this

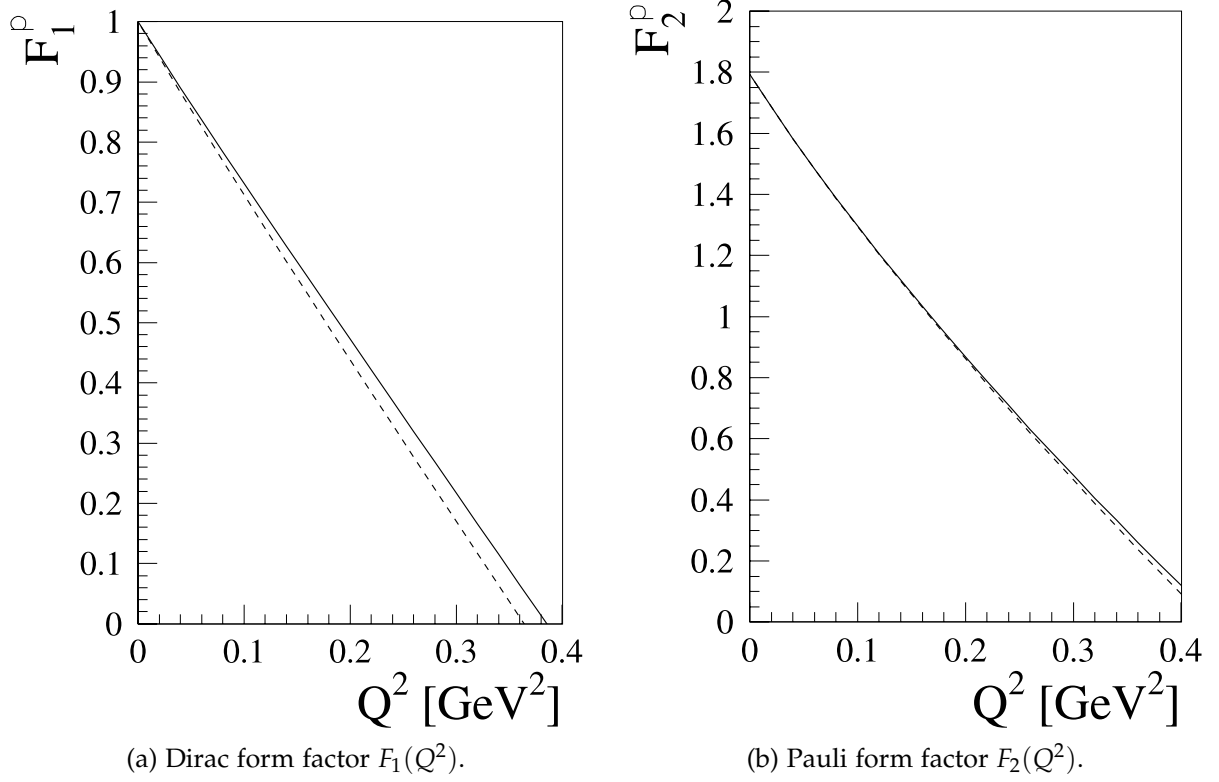


Figure 3.1: Dirac and Pauli form factors of the proton in relativistic chiral perturbation theory at $\mathcal{O}(q^4)$. The solid lines represent the extended on-mass-shell (EOMS) scheme, while the dashed lines show the result obtained using infrared regularization. Figures adapted from [XXX].

region determines the proton charge radius. Measurements at very small Q^2 , such as those performed by AMBER, are therefore especially sensitive to this quantity.

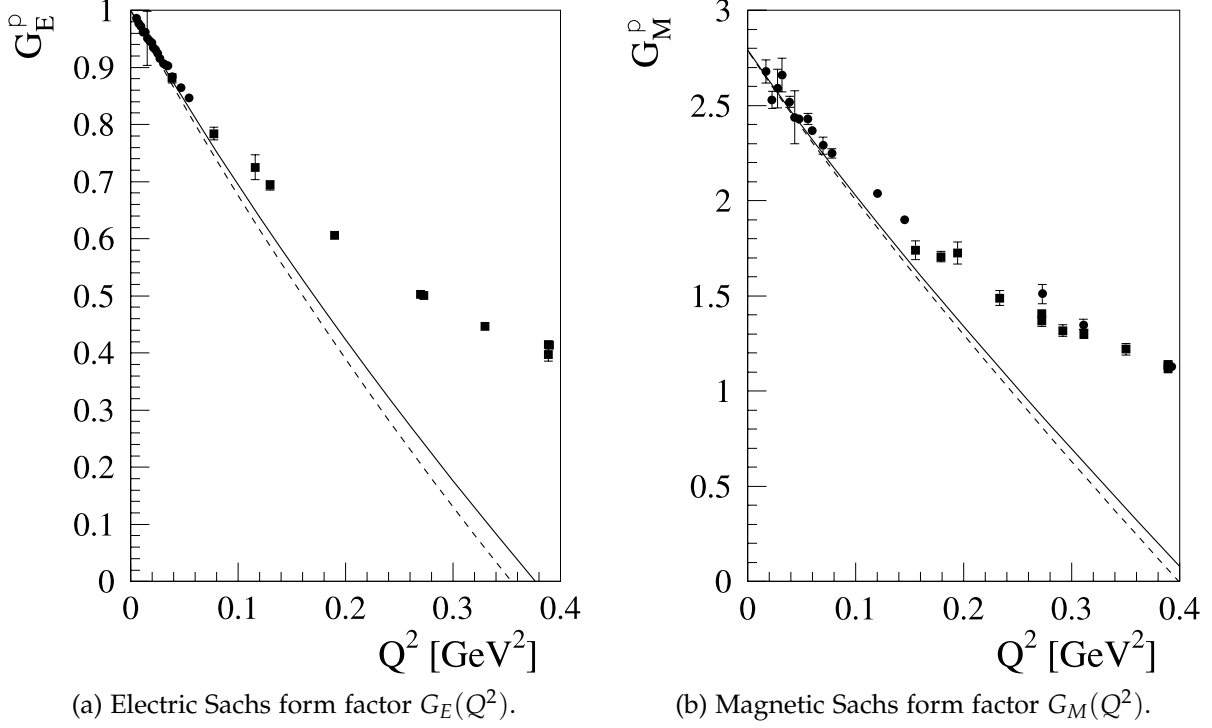


Figure 3.2: Electric and magnetic Sachs form factors of the proton at $\mathcal{O}(q^4)$. The solid lines represent the EOMS scheme and the dashed lines the infrared-regularization result. Figures adapted from [XXX].

The exchanged virtual photon connects the muon and proton vertices. Since it is an internal particle, it is represented by a propagator rather than an external photon wave function. In Feynman gauge, a convenient gauge choice in which the photon propagator takes a particularly simple form, it is given by

$$\frac{-ig_{\mu\nu}}{q^2}.$$

The muon and proton vertices carry the Lorentz indices μ and ν , respectively, corresponding to the leptonic and hadronic spinor chains. The virtual-photon propagator connects these two currents, with the metric tensor $g_{\mu\nu}$ contracting the Lorentz indices according to the Einstein summation convention.

Written with both Lorentz indices explicitly, the Born matrix element is

$$\mathcal{M}_{\text{Born}} = [Q_\mu \bar{u}(p_4)(-ie\gamma^\mu)u(p_2)] \frac{-ig_{\mu\nu}}{q^2} [\bar{u}(p_3)(-ie\Gamma^\nu(Q^2))u(p_1)]. \quad (3.27)$$

The contraction with the metric tensor lowers the Lorentz index of the proton vertex,

$$g_{\mu\nu}\Gamma^\nu(Q^2) = \Gamma_\mu(Q^2),$$

so that the matrix element can be written more compactly as

$$\mathcal{M}_{\text{Born}} = Q_\mu \frac{e^2}{q^2} \bar{u}(p_4)\gamma^\mu u(p_2) \bar{u}(p_3)\Gamma_\mu(Q^2)u(p_1). \quad (3.28)$$

The matrix element itself is not directly observable. The differential cross section depends on its squared modulus, summed over the unobserved final-state spins and averaged over the initial-state spins. For unpolarized muon–proton scattering,

$$\overline{|\mathcal{M}|^2} = \frac{1}{4} \sum_{s_1, s_2, s_3, s_4} |\mathcal{M}|^2, \quad (3.29)$$

where the factor 1/4 averages over the two possible spin states of each incoming spin-1/2 particle.

Writing the spin labels explicitly,

$$\mathcal{M} = Q_\mu \frac{e^2}{q^2} [\bar{u}(p_4, s_4)\gamma^\mu u(p_2, s_2)] [\bar{u}(p_3, s_3)\Gamma_\mu(Q^2)u(p_1, s_1)].$$

Its complex conjugate is

$$\mathcal{M}^* = Q_\mu \frac{e^2}{q^2} [\bar{u}(p_2, s_2)\gamma^\nu u(p_4, s_4)] [\bar{u}(p_1, s_1)\bar{\Gamma}_\nu(Q^2)u(p_3, s_3)],$$

where

$$\bar{\Gamma}_\nu = \gamma^0 \Gamma_\nu^\dagger \gamma^0.$$

Since $Q_\mu = -1$, the charge factor satisfies $Q_\mu^2 = 1$ in the squared amplitude. Thus,

$$|\mathcal{M}|^2 = \frac{e^4}{(q^2)^2} [\bar{u}(p_4)\gamma^\mu u(p_2)] [\bar{u}(p_2)\gamma^\nu u(p_4)] [\bar{u}(p_3)\Gamma_\mu u(p_1)] [\bar{u}(p_1)\bar{\Gamma}_\nu u(p_3)].$$

To evaluate the spin sums, the completeness relation for Dirac spinors is used,

$$\sum_s u(p, s)\bar{u}(p, s) = \not{p} + m,$$

where

$$\not{p} = \gamma^\mu p_\mu.$$

This relation converts the products of spinors into traces over gamma matrices. The spin-averaged squared matrix element can consequently be written as

$$\overline{|\mathcal{M}|^2} = \frac{e^4}{4(q^2)^2} L^{\mu\nu} H_{\mu\nu}, \quad (3.30)$$

where the leptonic tensor is

$$L^{\mu\nu} = \text{Tr} [(\not{p}_4 + m_\mu)\gamma^\mu(\not{p}_2 + m_\mu)\gamma^\nu],$$

and the hadronic tensor is

$$H_{\mu\nu} = \text{Tr} [(\not{p}_3 + m_p)\Gamma_\mu(Q^2)(\not{p}_1 + m_p)\bar{\Gamma}_\nu(Q^2)].$$

The factorization into a leptonic and a hadronic tensor separates the two parts of the interaction. The leptonic tensor describes the coupling of the virtual photon to the point-like muon, whereas the hadronic tensor contains the electromagnetic structure of the proton through its form factors.

For the muon, the leptonic tensor can be evaluated using the standard gamma-matrix trace identities,

$$L^{\mu\nu} = 4 \left[p_4^\mu p_2^\nu + p_4^\nu p_2^\mu - g^{\mu\nu} (p_2 \cdot p_4 - m_\mu^2) \right]. \quad (3.31)$$

The leptonic tensor satisfies current conservation and is therefore transverse to the transferred four-momentum,

$$q_\mu L^{\mu\nu} = q_\nu L^{\mu\nu} = 0.$$

To evaluate the hadronic tensor, it is convenient to introduce

$$\tau = \frac{Q^2}{4m_p^2} \quad (3.32)$$

and

$$P^\mu = p_1^\mu + p_3^\mu.$$

The on-shell conditions imply

$$P \cdot q = 0, \quad P^2 = 4m_p^2 + Q^2.$$

Expressed in terms of the Sachs form factors, the hadronic tensor takes the form

$$H_{\mu\nu} = 2 \frac{G_E^2(Q^2) + \tau G_M^2(Q^2)}{1 + \tau} P_\mu P_\nu - 2G_M^2(Q^2) (q_\mu q_\nu + Q^2 g_{\mu\nu}). \quad (3.33)$$

Because of current conservation, terms proportional to $q_\mu q_\nu$ do not contribute when the hadronic tensor is contracted with the leptonic tensor. Defining

$$A(Q^2) = \frac{G_E^2(Q^2) + \tau G_M^2(Q^2)}{1 + \tau}, \quad (3.34)$$

the contraction of Equation 3.31 and Equation 3.33 gives the exact spin-averaged Born matrix element,

$$\begin{aligned} |\overline{\mathcal{M}_{\text{Born}}}|^2 = \frac{e^4}{Q^4} \Big\{ & A(Q^2) \left[(s-u)^2 - Q^2 (4m_p^2 + Q^2) \right] \\ & + 2G_M^2(Q^2) Q^2 (Q^2 - 2m_\mu^2) \Big\}. \end{aligned} \quad (3.35)$$

Inserting Equation 3.35 into Equation 3.23 yields the exact laboratory-frame Born cross section,

$$\begin{aligned} \frac{d\sigma_{\text{Born}}}{d\Omega} = & \frac{1}{64\pi^2 m_p p_\mu} \frac{(p'_\mu)^2}{p'_\mu (m_p + E_\mu) - p_\mu E'_\mu \cos \theta_\mu} \frac{e^4}{Q^4} \\ & \times \left\{ A(Q^2) \left[(s-u)^2 - Q^2 (4m_p^2 + Q^2) \right] + 2G_M^2(Q^2) Q^2 (Q^2 - 2m_\mu^2) \right\}. \end{aligned} \quad (3.36)$$

3.4.1 Ultrarelativistic Limit

For the 100 GeV muon beam used in the AMBER proton-radius measurement, the incoming muon energy is much larger than the muon mass. It is therefore useful to consider the ultrarelativistic limit

$$E_\mu \gg m_\mu.$$

In this limit,

$$p_\mu \simeq E_\mu, \quad p'_\mu \simeq E'_\mu,$$

and the outgoing-muon energy becomes

$$E'_\mu = \frac{E_\mu}{1 + \frac{2E_\mu}{m_p} \sin^2 \left(\frac{\theta_\mu}{2} \right)}. \quad (3.37)$$

The momentum transfer can then be written as

$$Q^2 = 4E_\mu E'_\mu \sin^2 \left(\frac{\theta_\mu}{2} \right),$$

and differentiation with respect to the scattering angle gives

$$\frac{dQ^2}{d\theta_\mu} = \frac{2E_\mu^2 \sin \theta_\mu}{\left[1 + \frac{2E_\mu}{m_p} \sin^2 \left(\frac{\theta_\mu}{2} \right) \right]^2}. \quad (3.38)$$

The exact phase-space expression in Equation 3.23 correspondingly simplifies to

$$\frac{d\sigma}{d\Omega} = \frac{|\overline{\mathcal{M}}|^2}{64\pi^2 m_p^2 \left[1 + \frac{2E_\mu}{m_p} \sin^2 \left(\frac{\theta_\mu}{2} \right) \right]^2}. \quad (3.39)$$

The muon-mass terms also disappear from the leptonic tensor,

$$L^{\mu\nu} = 4 (p_4^\mu p_2^\nu + p_4^\nu p_2^\mu - g^{\mu\nu} p_4 \cdot p_2) .$$

The hadronic tensor in Equation 3.33, however, remains unchanged, since the ultrarelativistic approximation concerns the muon kinematics rather than the proton structure.

Combining the leptonic and hadronic tensors leads to the Rosenbluth form of the elastic Born cross section,

$$\frac{d\sigma_{\text{Born}}}{d\Omega} = \frac{d\sigma_{\text{Mott}}}{d\Omega} \frac{\tau G_M^2(Q^2) + \epsilon G_E^2(Q^2)}{\epsilon(1 + \tau)} . \quad (3.40)$$

Here, the Mott factor describes scattering from a point-like charged target. With the recoil factor included in the convention used here, it is

$$\frac{d\sigma_{\text{Mott}}}{d\Omega} = \frac{\alpha^2 \cos^2(\theta_\mu/2)}{4E_\mu^2 \sin^4(\theta_\mu/2)} \frac{E'_\mu}{E_\mu} , \quad (3.41)$$

where

$$\alpha = \frac{e^2}{4\pi}$$

is the fine-structure constant. The virtual-photon polarization parameter is

$$\epsilon = \left[1 + 2(1 + \tau) \tan^2\left(\frac{\theta_\mu}{2}\right) \right]^{-1} . \quad (3.42)$$

The Rosenbluth expression makes the role of the proton structure explicit: the electric and magnetic contributions enter separately through G_E and G_M , while a point-like proton would correspond to structure-independent form factors.

For the numerical description used later in this thesis, the Sachs form factors are represented by the empirical dipole parametrization

$$G_D(Q^2) = \left(1 + \frac{Q^2}{\Lambda^2} \right)^{-2} , \quad (3.43)$$

with

$$\Lambda^2 = 0.71 \text{ GeV}^2 .$$

The proton electric and magnetic form factors are then parametrized as

$$G_E(Q^2) = G_D(Q^2) = \left(1 + \frac{Q^2}{0.71 \text{ GeV}^2} \right)^{-2} , \quad (3.44)$$

and

$$G_M(Q^2) = \mu_p G_D(Q^2) = \mu_p \left(1 + \frac{Q^2}{0.71 \text{ GeV}^2} \right)^{-2} . \quad (3.45)$$

3.5 Radiative Corrections

This part is still under construction.

3.5.1 Notes

The Born approximation provides the leading contribution to elastic muon–proton scattering, but a precision determination of the proton charge radius requires higher-order QED effects to be included. In this thesis, radiative corrections are considered through next-to-leading order (NLO).

The Born amplitude is of order e^2 , corresponding to a cross section of order α^2 . At NLO, the cross section receives corrections of order α^3 from two different types of contributions: virtual one-loop corrections and real-photon emission. The corresponding processes are illustrated in Figure 3.3.

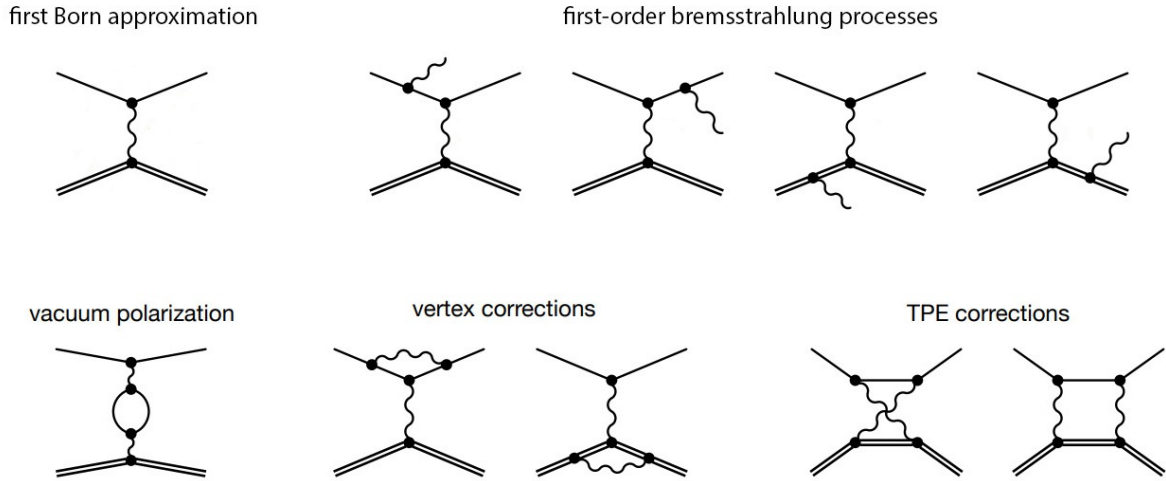


Figure 3.3: Leading-order and next-to-leading-order contributions to muon–proton scattering. At leading order, the interaction proceeds through the exchange of a single virtual photon. At NLO, virtual loop corrections and real-photon emission contribute. Figure taken from [M.Depour].

The figure illustrates the qualitative difference between the Born and NLO contributions. While the Born process contains only the outgoing muon and proton, real-photon emission changes the final-state topology to a three-particle state. This is experimentally relevant because the emitted photon carries away part of the energy and momentum and therefore changes the reconstructed kinematics of the muon–proton system.

Virtual Corrections Virtual corrections arise from one-loop diagrams. They include self-energy, vertex, vacuum-polarization, and box contributions such as two-photon exchange. Since the virtual diagrams lead to the same external final state as the Born process, their

contribution to the NLO cross section enters through the interference between the Born and one-loop amplitudes,

$$2 \operatorname{Re} (\mathcal{M}_{\text{Born}}^* \mathcal{M}_{\text{virtual}}).$$

Ultraviolet divergences appearing in the loop integrals are removed through renormalization. Infrared divergences, however, cancel only when the virtual contribution is combined consistently with the corresponding real-emission contribution for an infrared-safe observable.

Real Photon Emission The real NLO contribution is the bremsstrahlung process

$$p\mu \rightarrow p\mu\gamma, \tag{3.46}$$

in which an additional photon is emitted from one of the charged external particles. In contrast to the non-radiative $2 \rightarrow 2$ process, real photon emission produces a three-body final state. The photon carries energy and momentum away from the muon–proton system and can therefore significantly alter observables such as the outgoing-muon energy, scattering angle, and reconstructed momentum transfer.

The virtual and real contributions must be combined using the same definition of the measured observable. Their sum is infrared finite, while the resulting numerical correction depends on the experimental phase-space definition, acceptance requirements, and photon-energy cuts. This makes a fully differential treatment of radiative corrections particularly important for the AMBER proton-radius measurement.

3.5.2 Structure of the NLO Cross Section

3.5.3 Virtual Corrections

3.5.4 Real Photon Emission

3.5.5 Infrared Cancellation

4 Simulation with McMule

5 Results

6 Discussion

7 Conclusion and Outlook

Diese Arbeit war ganz toll.

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